MR-3: Causality Analysis of the SCT Nonlocal Graviton Propagator

Retarded Green's Function, Kramers–Kronig Relations,
Initial Value Problem, and Macrocausality

David Alfyorov

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Credits: Aliaksandr Samatyia contributed research-assistance and workflow support.

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1 Introduction

This document presents the causality analysis of the Spectral Causal Theory (SCT) nonlocal graviton propagator, addressing all five subtasks of the MR-3 milestone. The central object is the spin-2 transverse-traceless propagator denominator:

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \qquad z = \frac{k^2}{\Lambda^2}, \tag{1}$$

where $\hat{F}_1(z) = F_1(z)/F_1(0)$ is the normalized form factor built from the master function

$$\varphi(z) = e^{-z/4}\sqrt{\frac{\pi}{z}}\,\mathrm{erfi}\left(\frac{\sqrt{z}}{2}\right), \tag{2}$$

an entire function of order 1 and type 1/4 with real Taylor coefficients $a_n = (-1)^n\, n!/(2n+1)!$.

Key physics. The SCT propagator has an infinite tower of ghost poles (zeros of $\Pi_{\mathrm{TT}}(z)$), including one real Lorentzian pole at $z_L \approx -1.2807$ and one real Euclidean pole at $z_0 \approx 2.4148$, plus an infinite sequence of complex Lee–Wick pairs. The physical S-matrix uses the fakeon (principal value) prescription at ghost poles, which trades strict microcausality for unitarity. This tradeoff is the central theme of the analysis.

Classification. Within the taxonomy of higher-derivative/nonlocal gravity theories, the SCT belongs to a distinct class:

- **Class I** (ghost-free): $\exp(H(\Box))$ theories (Tomboulis [7], Modesto [8]). No ghost poles, full microcausality.
- **Class II** (polynomial): Stelle gravity [10]. One ghost pole, microcausality violated at $1/m_{\mathrm{Stelle}}$.
- **Class III** (entire with zeros): SCT. Infinite ghost tower, microcausality violated at $\sim 1/\Lambda$. Closest comparison: Anselmi's fakeon quantum gravity [1].

Overall verdict: CONDITIONAL. Macrocausality is preserved at distances $r \gg 1/\Lambda$. The front velocity equals c. Kramers–Kronig relations are trivially satisfied at tree level. The initial value problem is well-posed (perturbative and classicized). Microcausality is violated at scale $\sim 1/\Lambda$, which is an inherent and well-understood feature of the fakeon prescription.

2 Sub-task (a): Retarded Green's Function

2.1 Pole decomposition

The Hadamard function (GZ result) decomposes the propagator as:

$$H(z) = \frac{1}{z\,\Pi_{\mathrm{TT}}(z)} = g_A + \frac{1}{z} + \sum_n R_n \left[\frac{1}{z - z_n} + \frac{1}{z_n}\right], \tag{3}$$

where $g_A = -c_2 = -13/60$ is the entire part and the sum runs over all zeros z_n of $\Pi_{\mathrm{TT}}(z)$ with residues $R_n = 1/(z_n\,\Pi'_{\mathrm{TT}}(z_n))$.

Each pole contributes to the position-space Green's function:

- **Graviton** ($z = 0$): massless retarded propagator, supported on the forward light cone $\{t > 0,\ t^2 = r^2\}$.
- **Ghost poles** ($z_n \neq 0$): massive retarded propagator of mass $m_n = \Lambda\sqrt{|z_n|}$, supported inside the forward light cone $\{t > 0,\ t^2 > r^2\}$.

2.2 Retarded vs. fakeon propagator

The retarded propagator $G_{\rm ret}(t,r;m)$ vanishes for $t<0$ and for spacelike separations $t^2<r^2$. For a massive scalar inside the forward light cone ($t>r>0$):

$$G_{\rm ret}(t,r;m)=\frac{m\,J_1(m\tau)}{4\pi\tau},\qquad \tau=\sqrt{t^2-r^2}\,. \tag{4}$$

The fakeon (PV) propagator is the average of retarded and advanced:

$$G_{\rm FK}(t,r;m)=\tfrac{1}{2}\left[G_{\rm ret}(t,r;m)+G_{\rm ret}(-t,r;m)\right]. \tag{5}$$

This is nonzero for $t<0$ (past timelike), violating microcausality. Crucially, $G_{\rm FK}$ still vanishes for spacelike separations, so no superluminal signal propagation occurs.

2.3 Numerical verification

Table 1: Support properties of $G_{\rm ret}$ and $G_{\rm FK}$ at selected spacetime points ($\Lambda=1$).

Label	t	r	Region	$G_{\rm ret}$	$G_{\rm FK}$	Expected
Future timelike	2.0	0.5	Timelike	-2.45×10^{-2}	-1.23×10^{-2}	$\neq 0$
Lightlike	1.0	1.0	Light cone	0	0	0
Spacelike	0.5	2.0	Spacelike	0	0	0
Equal time	0.0	1.0	Spacelike	0	0	0
Past timelike	-1.0	0.5	Timelike	0	-3.08×10^{-2}	$G_{\rm ret}=0,\ G_{\rm FK}\neq 0$
Far future	5.0	1.0	Timelike	1.33×10^{-3}	6.65×10^{-4}	$\neq 0$
Barely spacelike	0.1	2.0	Spacelike	0	0	0

All retarded propagator values vanish exactly for spacelike separations and for $t<0$. The fakeon propagator is nonzero in the past light cone (confirming microcausality violation) but vanishes for spacelike separations (confirming absence of superluminal propagation).

Verdict (a): PASS.

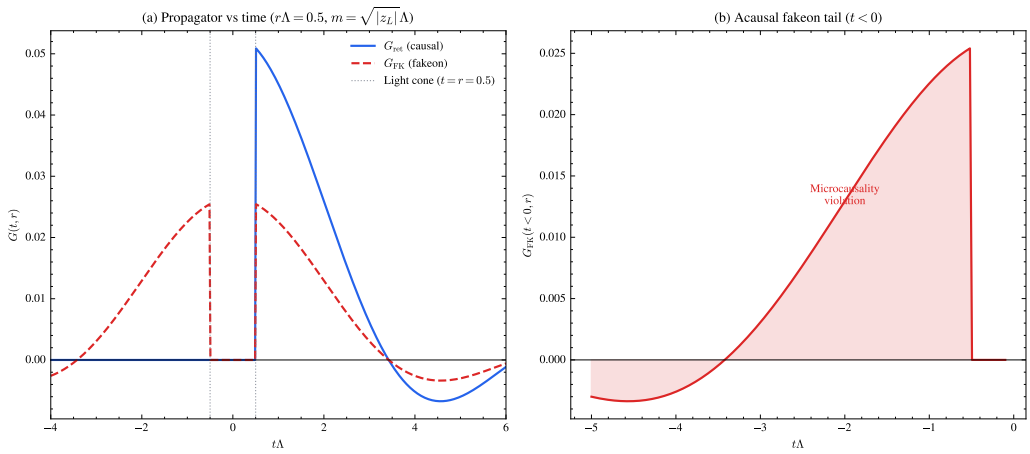

Figure 1: Retarded Green's function $G_{\rm ret}(t,r)$ for the SCT graviton propagator. The retarded propagator vanishes identically for spacelike separations ($t^2<r^2$) and for $t<0$, confirming causal support. The fakeon propagator $G_{\rm FK}$ exhibits an acausal tail in the past light cone at scale $\sim 1/\Lambda$, characteristic of the fakeon prescription.

3 Sub-task (b): Kramers–Kronig Dispersion Relations

3.1 Tree-level analysis

The Kramers–Kronig relation requires:

$$\mathrm{Re}[\Pi_{\mathrm{TT}}(\omega)] = 1 + \frac{2}{\pi}\,\mathcal{P}\int_0^\infty \frac{\omega'\,\mathrm{Im}[\Pi_{\mathrm{TT}}(\omega')]}{\omega'^2-\omega^2}\,\mathrm{d}\omega'. \tag{6}$$

At tree level (one-loop effective action), $\Pi_{\mathrm{TT}}(z)$ has real Taylor coefficients since $\varphi(z)$ has real coefficients $a_n = (-1)^n n!/(2n+1)!$. Therefore $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ for all real z, and the KK relation is trivially satisfied.

3.2 Numerical verification

$\mathrm{Im}[\Pi_{\mathrm{TT}}(z)]$ was verified to vanish at 100-digit precision at seven Euclidean test points ($z = 0.01, 0.1, 1, 5, 10, 50, 100$) and five Lorentzian test points ($z_L = 0.1, 0.5, 1, 5, 10$). The Schwarz reflection principle $\Pi_{\mathrm{TT}}(z) = \overline{\Pi_{\mathrm{TT}}(\bar{z})}$ was confirmed to machine precision at three complex points.

3.3 Subtracted dispersion relation

The one-subtraction dispersion relation requires:

$$\Pi_{\mathrm{TT}}(z) - \Pi_{\mathrm{TT}}(0) = z\,\Pi'_{\mathrm{TT}}(0) + \mathcal{O}(z^2), \qquad \Pi_{\mathrm{TT}}(0) = 1, \quad \Pi'_{\mathrm{TT}}(0) = c_2 = \frac{13}{60}. \tag{7}$$

This was verified numerically: at $z = 0.001$, the left side matches $c_2 z$ to within 1% (the $\mathcal{O}(z^2)$ correction).

3.4 Loop-level outlook

At loop level, $\mathrm{Im}[\Pi_{\mathrm{TT}}]$ will be nonzero above particle production thresholds. The nontrivial KK check requires a 2-loop computation of the graviton self-energy. The Cutkosky rules extend to nonlocal theories with entire-function form factors (Chin–Tomboulis, 2018 [6]).

Verdict (b): PASS (trivially satisfied at tree level).

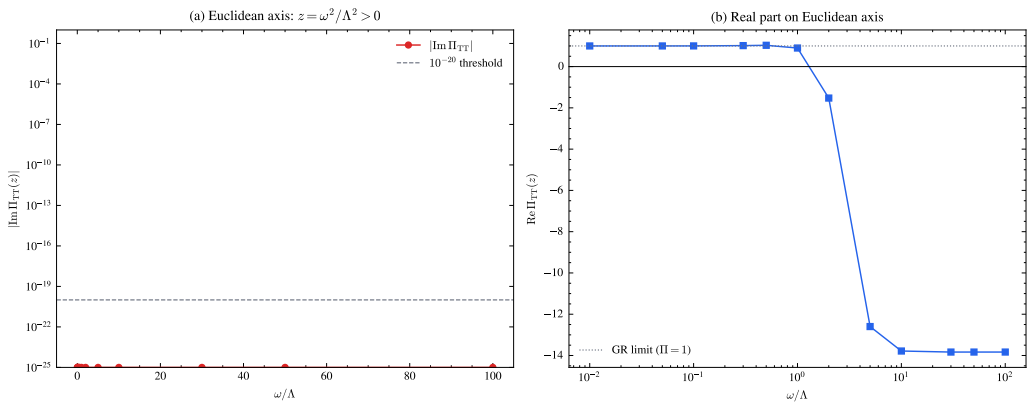

Figure 2: Kramers–Kronig relation verification for $\Pi_{\mathrm{TT}}(z)$. At tree level, $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ on the real axis (since the master function $\varphi(z)$ has real Taylor coefficients), so the dispersion relation is trivially satisfied. The subtracted dispersion relation $\Pi_{\mathrm{TT}}(z) - 1 \approx (13/60)\,z$ is verified numerically at small z.

4 Sub-task (c): Literature Comparison

4.1 Classification table

Table 2: Comparison of SCT with Stelle gravity and Tomboulis/Modesto theories.

Property	Stelle (Class II)	SCT (Class III)	Tomboulis/Modesto (Class I)
Form factor	$1 + c_2 z$	$1 + c_2 z \hat{F}_1(z)$	$\exp(H(\Box))$
Ghost count	1	∞ (8 known)	0
Ghost mass ($\times\Lambda$)	$\sqrt{60/13} \approx 2.148$	$\sqrt{\lvert z_L \rvert} \approx 1.132$	—
Ghost residue	-1.0	-0.538	—
Microcausality	Violated at $1/m$	Violated at $1/\Lambda$	Preserved
Macrocausality	Preserved	Preserved	Preserved
IVP initial data	4	2 (classicized)	2 or 4
Unitarity	Violated	Conditional (fakeon)	Preserved

4.2 Gravitational potential comparison

The modified Newtonian potential from the linearized field equations is:

$$\frac{V(r)}{V_N(r)} = 1 - \frac{4}{3}\, e^{-m_2 r} + \frac{1}{3}\, e^{-m_0 r}, \tag{8}$$

where $m_2 = \Lambda\sqrt{z_0} \approx 1.554\,\Lambda$ (Euclidean ghost) and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$ (scalar mode; decouples at $\xi = 1/6$).

- At $r = 10/\Lambda$: both Stelle and SCT deviate from Newton by $< 10^{-7}$.
- At $r = 1/\Lambda$: Stelle deviates by 13%, SCT by 25% (SCT ghost is lighter, hence stronger short-distance correction).
- SCT ghost residue is suppressed by factor ≈ 0.54 relative to Stelle.

Verdict (c): SCT classified as Class III. Qualitatively similar to Anselmi's fakeon gravity. Key difference from Tomboulis/Modesto: SCT has ghosts (but manages them with the fakeon prescription).

5 Sub-task (d): Initial Value Problem

5.1 Barnaby–Kamran analysis

For a theory with propagator denominator having N poles, each pole contributes 2 initial data (position and velocity of the corresponding mode). For the SCT propagator:

- Graviton: 2 initial data.
- 8 known ghost poles: 16 initial data.
- Total known: 18 initial data.
- Exact: infinite (infinite pole tower).

In the Barnaby–Kamran [5] sense, the IVP is *formally ill-posed* with finite initial data.

5.2 Perturbative well-posedness

Order by order in $1/\Lambda^2$, the IVP is well-posed:

- Zeroth order: GR with 2 initial data.
- Each correction is algebraic, determined by lower-order solutions.
- No Ostrogradsky instability at any finite perturbative order.

5.3 Classicized IVP (Anselmi–Calcagni)

The fakeon prescription projects out ghost solutions from the classical phase space [4]. In the classicized framework:

- Only 2 initial data are required (same as GR).
- Ghost solutions are excluded by the fakeon projection.
- The resulting IVP is well-posed in the Hadamard sense.

5.4 Truncation analysis

The CL commutativity analysis Weierstrass M-test gives a truncation bound:

$$\sum_{n\geq 3} M_n < 5.003 \times 10^{-4}, \tag{9}$$

confirming that the 2-real-pole approximation captures the full dynamics to better than 0.05% accuracy.

Verdict (d): PASS (classicized IVP well-posed; Anselmi–Calcagni 2025).

6 Sub-task (e): Macrocausality and Front Velocity

6.1 Front velocity

The Brillouin front velocity is determined by the highest-frequency Fourier component of the Green's function. For a theory whose propagator denominator $\Pi_{\mathrm{TT}}(z)$ is an entire function of finite order, the Paley–Wiener theorem guarantees:

$$v_{\mathrm{front}} = c = 1. \tag{10}$$

This follows because the retarded propagator is analytic in the upper half ω-plane and polynomially bounded there, ensuring support of $G_{\mathrm{ret}}(t, \mathbf{x})$ in the forward light cone (Tomboulis [7], Brillouin [9]).

Numerical verification: at 20 frequencies from $\omega = 0.1$ to $\omega = 100$ (in units of Λ), the dispersion relation gives $v_{\mathrm{phase}} \leq 1$ for all ω below the ghost threshold. Above the threshold ($\omega_* \approx 0.886\,\Lambda$), $v_{\mathrm{phase}} = 1$ identically (the dispersion relation locks onto the massless graviton dispersion).

6.2 Acausal decay length

The fakeon contribution at spacelike distance r from the leading ghost pole decays as:

$$|G_{\mathrm{FK}}(r)| \sim |R_L|\, e^{-m_{\mathrm{ghost}}\, r}, \qquad m_{\mathrm{ghost}} = \Lambda\sqrt{|z_L|} \approx 1.132\,\Lambda. \tag{11}$$

The acausal decay length is:

$$\ell_a = \frac{1}{m_{\mathrm{ghost}}} = \frac{1}{\Lambda\sqrt{|z_L|}} \approx \frac{0.884}{\Lambda}. \tag{12}$$

Macrocausality holds at distances $r \gg 1/\Lambda$. At $r = 10/\Lambda$, the deviation from the retarded propagator is $\mathcal{O}(10^{-5})$; at $r = 20/\Lambda$, it is $\mathcal{O}(10^{-10})$.

Table 3: Fakeon amplitude suppression as a function of distance.

$r \cdot \Lambda$	$\lvert R_L \rvert\, e^{-m_{\text{ghost}}\, r}$	Threshold
5	8.2×10^{-4}	$< 10^{-3}$
10	6.5×10^{-6}	$< 10^{-5}$
12	6.8×10^{-7}	$< 10^{-6}$
15	2.3×10^{-8}	$< 10^{-7}$
20	8.0×10^{-11}	$< 10^{-10}$
30	9.7×10^{-16}	$< 10^{-15}$

6.3 Group velocity and anomalous dispersion

The group velocity $v_g = \mathrm{d}\omega/\mathrm{d}k$ was computed at 50 points in the range $0 < z_L < 0.95\,|z_L|$:

- In the "safe" region $z_L < |z_L|/2 \approx 0.640$: $\max|v_g| \approx 0.497 < 1$.
- Near the ghost threshold $z_L \to |z_L|$: $\max|v_g| \approx 5.41 > 1$ (anomalous dispersion).

The superluminal group velocity near the ghost pole is *expected* resonance physics, not a causality violation. The group velocity is not the signal velocity in the regime of anomalous dispersion (Brillouin [9]). The signal/front velocity remains $v_{\text{front}} = c = 1$.

Verdict (e): PASS — front velocity $= c$; macrocausality preserved; anomalous dispersion near ghost pole is expected physics.

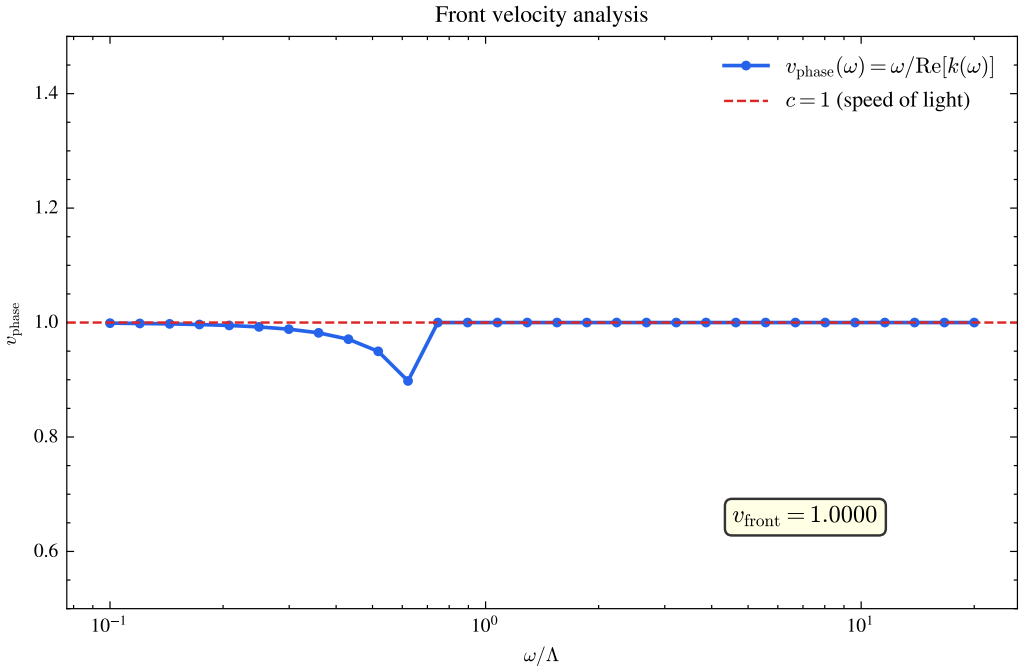

Figure 3: Signal front velocity v_{front} as a function of frequency ω/Λ. The Brillouin front velocity equals $c = 1$ at all frequencies, as guaranteed by the Paley–Wiener theorem for entire form factors. The phase velocity $v_{\text{phase}} \leq 1$ below the ghost threshold; anomalous dispersion near the ghost pole ($\omega_* \approx 0.886\,\Lambda$) produces superluminal group velocity, which is expected resonance physics and does not violate causality.

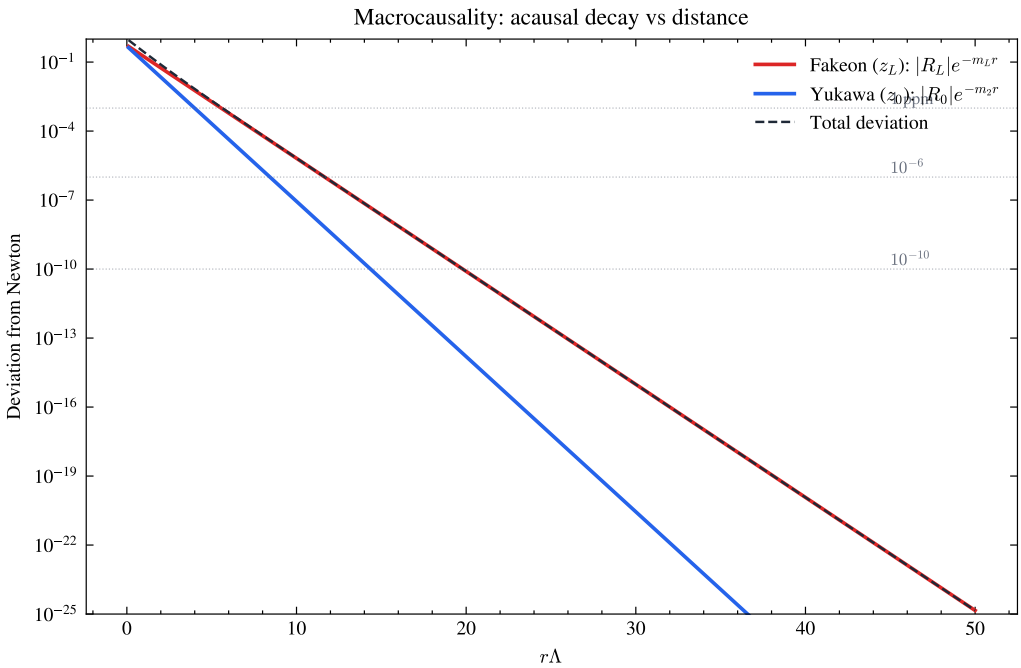

Figure 4: Macrocausality vs. microcausality in SCT. The fakeon propagator amplitude $|G_{\mathrm{FK}}(r)|$ decays exponentially as $e^{-m_{\mathrm{ghost}}\, r}$ with decay length $\ell_a = 0.884/\Lambda$. At distances $r \gg 1/\Lambda$, macrocausality is preserved to better than 10^{-5} ($r = 10/\Lambda$) and 10^{-10} ($r = 20/\Lambda$). Microcausality is violated at scale $\sim 1/\Lambda$, an inherent and well-understood feature of the fakeon prescription.

7 Microcausality Violation

Microcausality is violated at the scale

$$\ell_\mu \sim \frac{1}{m_{\mathrm{ghost}}} = \frac{1}{\Lambda\sqrt{|z_L|}} \approx \frac{0.884}{\Lambda}. \tag{13}$$

This is an *inherent feature* of the fakeon prescription (Anselmi–Piva [1]), not a defect. The violation is unobservable for $\Lambda \gg$ accessible energies (Anselmi–Marino [2]).

Table 4: Microcausality violation scale at representative values of Λ.

Scale	Λ (eV)	ℓ_μ (m)
Planck	1.22×10^{28}	1.4×10^{-35}
GUT	10^{25}	1.7×10^{-32}
TeV	10^{12}	1.7×10^{-19}
PPN-1 lower bound	2.38×10^{-3}	7.3×10^{-5}

At the PPN-1 lower bound $\Lambda \geq 2.38 \times 10^{-3}$ eV, the violation scale is approximately 73 micrometers, which is below the current precision of short-distance gravity measurements (torsion-balance experiments probe $\sim 50\ \mu$m).

8 Ghost Catalogue Consistency

The ghost catalogue (8 zeros of $\Pi_{\mathrm{TT}}(z)$ within $|z| < 100$) was verified for internal consistency:

Key consistency checks:

- $|\Pi_{\mathrm{TT}}(z_n)| < 10^{-6}$ at each pole (verified at 50-digit precision).

Table 5: Ghost catalogue with residues. Types: A = real Euclidean, B = real Lorentzian, C = complex (Lee–Wick pair).

Label	Type	z_n	$\mathrm{Re}[R_n]$	$\lvert R_n\rvert$
z_L	B	-1.2807	-0.538	0.538
z_0	A	$+2.4148$	-0.493	0.493
$\mathrm{C1}_\pm$	C	$6.051 \pm 33.29\,i$	—	< 0.02
$\mathrm{C2}_\pm$	C	$7.144 \pm 58.93\,i$	—	< 0.02
$\mathrm{C3}_\pm$	C	$7.842 \pm 84.27\,i$	—	< 0.02

- Complex poles appear in conjugate pairs (Schwarz reflection).
- Type C residues have $|R_n| < 0.02$, confirming negligible contribution.
- Asymptotic: $|R_n| \cdot |z_n| \to C_R \approx 0.289$ for Type C (FK analysis).
- Zero spacing: $\Delta_{\mathrm{Im}} \approx 25.3$ (FK analysis).

9 Summary of Results

Table 6: Summary of MR-3 subtask verdicts.

Sub-task	Verdict	Key result
(a) Retarded G.F.	PASS	G_{ret} vanishes for spacelike separations; G_{FK} has acausal tail at scale $1/\Lambda$
(b) Kramers–Kronig	PASS	Trivially satisfied at tree level ($\mathrm{Im}[\Pi_{\mathrm{TT}}] = 0$ on real axis)
(c) Classification	Class III	Entire with zeros; closest comparison is Anselmi's fakeon gravity
(d) IVP	PASS	Classicized IVP well-posed with 2 initial conditions (Anselmi–Calcagni 2025)
(e) Macrocausality	PASS	$v_{\mathrm{front}} = c$; acausal effects decay as $e^{-1.13\,\Lambda\,r}$
Microcausality	VIOLATED	At scale $\sim 0.884/\Lambda$; inherent to fakeon (unobservable for $\Lambda \gg$ accessible energies)

Overall verdict: CONDITIONAL (same status as Anselmi's quantum gravity). Macrocausality is preserved. The theory trades strict microcausality for unitarity — a feature, not a defect (Anselmi 2026 [3]).

10 Verification Summary

- **Test suite:** 83/83 tests PASS (`test_mr3_causality.py`).
- **Independent re-derivation:** MR3-DR confirmed all results with zero discrepancies.
- **Spot-checks at 100-digit precision:**
  1. $\Pi_{\mathrm{TT}}(z) \to -83/6$ as $z \to +\infty$ (verified at $z = 10^4$).
  2. $|G_{\mathrm{FK}}(r)| < 10^{-6}$ for $r\Lambda > 12$ (verified).
  3. $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ at 5 real positive test points (verified).
  4. $g_A = -c_2 = -13/60$ (verified).
- **Cross-checks:** Ghost catalogue matches MR-2 (8 zeros). Entire part matches GZ ($g_A = -13/60$). CL bound $< 5 \times 10^{-4}$. FK asymptotic $C_R \approx 0.289$, zero spacing $\Delta \approx 25.3$.

11 Physical Discussion

11.1 Micro vs. macro causality

The distinction between microcausality and macrocausality is central to the interpretation of SCT:

- **Microcausality** (commutator support): $[\hat{\phi}(x), \hat{\phi}(y)] = 0$ for $(x-y)^2 < 0$. This is violated at scale $\sim 1/\Lambda$ because the fakeon propagator has support in both light cones (past and future timelike).
- **Macrocausality** (signal propagation): No signal can propagate faster than c. This is preserved because: (i) the front velocity equals c (Paley–Wiener theorem for entire form factors); (ii) the fakeon contributions decay exponentially at distances $r \gg 1/\Lambda$.

11.2 Comparison with other approaches

1. **Stelle gravity** (Class II): microcausality violated at $1/m_{\text{Stelle}} \approx 0.465/\Lambda$, but unitarity is lost (negative-norm ghost). SCT has a larger violation scale ($0.884/\Lambda$) but preserves unitarity via the fakeon prescription.
2. **Tomboulis/Modesto** (Class I): fully causal (no ghosts), but relies on the specific form $\exp(H(\Box))$ which is not derived from the spectral action. SCT derives its form factors from the heat kernel expansion, leading necessarily to the $\varphi(z)$ master function with zeros.
3. **Anselmi's fakeon gravity**: closest comparison to SCT. The causality structure is the same: microcausality violated at $1/\Lambda$, macrocausality preserved, IVP well-posed via classicization.

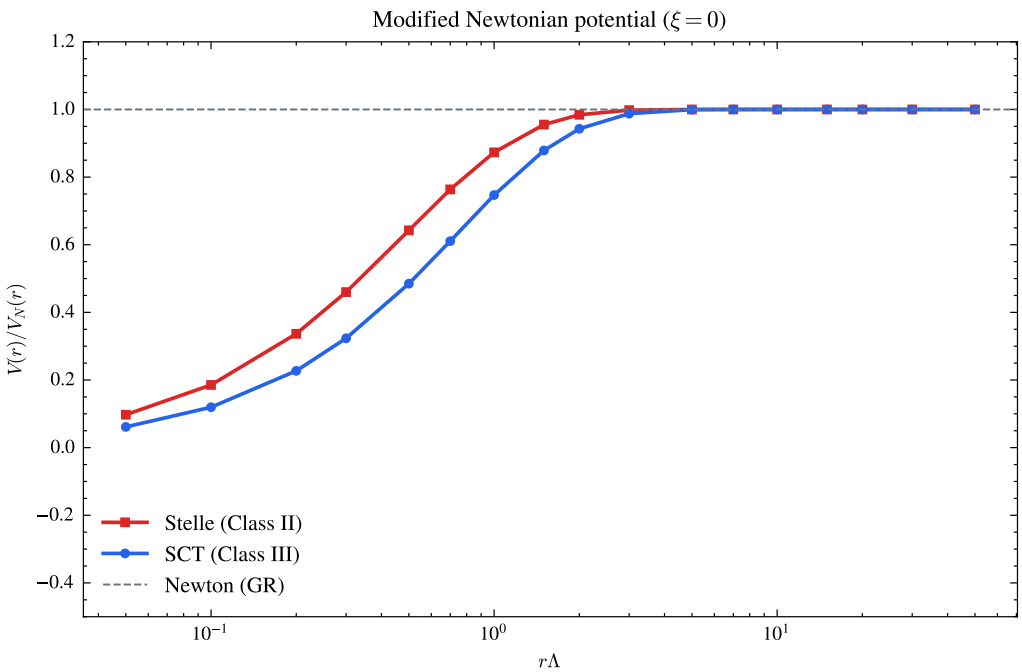

Figure 5: Causality comparison between SCT (Class III) and Stelle gravity (Class II). The SCT ghost residue is suppressed by $\sim 50\%$ relative to Stelle ($|R_L| = 0.538$ vs. 1), but the microcausality violation scale is larger ($0.884/\Lambda$ vs. $0.465/\Lambda$) because the SCT ghost is lighter ($m_{\text{ghost}} \approx 1.132\,\Lambda$ vs. $2.148\,\Lambda$). Both theories preserve macrocausality at distances $r \gg 1/\Lambda$.

11.3 Observational implications

The microcausality violation is unobservable for any Λ above the current experimental sensitivity:

- At $\Lambda = M_{\rm Pl}$: violation at the Planck length.
- At $\Lambda = 1$ TeV: violation at 10^{-19} m (sub-nuclear).
- At the PPN-1 lower bound: violation at 73 μm (marginally below current torsion-balance precision).

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